

AMENDMENT NO. 3 JULY 1979

TO

IS: 2825-1969 CODE FOR UNFIRED PRESSURE VESSELS

Alterations

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1. (Page 8, Table 1.1, Sl No. 4, col 2) — Substitute 'Type of joints' for 'Type of joints'.
2. (Page 8) — Add the following foot-note at the end of the page:

*These joints refer to all categories of Class 1 vessels and categories A and B only of Class 2 and 3 vessels. For categories C and D, details shown in Appendix G or equivalent shall be used, as applicable to the relevant class of the vessel.'
3. (Page 8, Table 1.1, Sl No. 4, col 3) — Substitute the following for the existing matter:
 - i) Double welded butt joints with full penetration excluding butt joints with metal backing strips which remain in place for categories A, B and C.
 - ii) Single welded butt joints with backing strip for categories B and C. $J = 0.9$ (see 6.3.6.1).
 - iii) Full penetration weld extending through the entire thickness of vessel wall or nozzle wall for category D.
4. (Page 10, clause 2.2.1) — Add the following matter at the end:

'However, for determining the allowable stress values for **ferrous material** for design temperatures (see 1.2.4) up to and including 400°C, the criteria relating to creep properties (that is, average stress to produce rupture in 100 000 hours and average stress to produce a total creep strain of one percent in 100 000 hours) listed in Table 2.1 shall not be taken into consideration.'
5. [Page 31, clause 3.8.5.2 (d) (1), equation (3.32)] — Substitute the following for the existing equation:

$$A = 0.5 d \cdot t_r \dots\dots\dots (3.32)'$$
6. (Page 71, clause 6.4.11, first sentence) — Substitute the following for the existing matter:

6.4.11 Plates Welded Prior to Forming — Seams in plates may be welded prior to forming provided they meet the specified non-destructive test requirements after forming.'
7. [Page 81, clause 7.1.3.4 (c)] — Substitute the following for the existing matter:

'c) if in arc welding of single welded butt joints a backing strip is omitted.'
8. [Page 83, clause 7.1.5 (d)] — Substitute the following for the existing matter:

'd) Three notched bar impact test specimens to be taken transversely to the weld (see 8.5.8) as near as possible to the face side of the last pass of the weld on outer plate surface. If plate thickness exceeds 40 mm, one more set of specimens shall be taken at about midway between centre of thickness and opposite side surface.'
9. [Page 97, clause 8.5.1.3 (d)] — Substitute the following for the existing matter:

'd) Three notched bar impact test specimens, to be taken transversely to the weld [see 7.1.5 (d)].'
10. (Pages 195 to 223, Figures in Appendix G) — Substitute 'permitted' for 'recommended' in various figures.
11. (Appendix G) — Incorporate a tell-tale hole in the following figures:

'Fig. G.6, Fig. G.7, Fig. G.12, Fig. G.26, Fig. G.27(A), Fig. G.27(B), Fig. G.29, Fig. G.30, Fig. G.31, Fig. G.32, Fig. G.33, Fig. G.34 and Fig. G.35.'
12. (Page 195, Fig. G.3) — Add the following note at the end:

'd) First sketch is permitted for Class 3 vessels only.'
13. [Page 197, Fig. G.6, note (b)] — Substitute the following notes for the existing note:

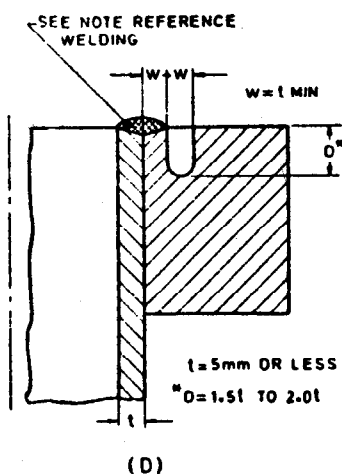
'b) Permitted for Class 3 vessels only.
 c) Special attention should be paid to the examination of the plate edges before and after welding.'
14. [Page 197, Fig. G.7, note (b)] — Substitute the following for the existing note:

'b) Permitted for Class 2 and Class 3 vessels only.'
15. [Page 197, Fig. G.8, note (b)] — Substitute the following notes for the existing note:

'b) Its use when thermal gradient may cause overstress in welds to be avoided. Permitted for Class 2 and Class 3 vessels only. The weld sizes are minimum.
 c) The pad is not to be taken into account in calculating the reinforcement required.'

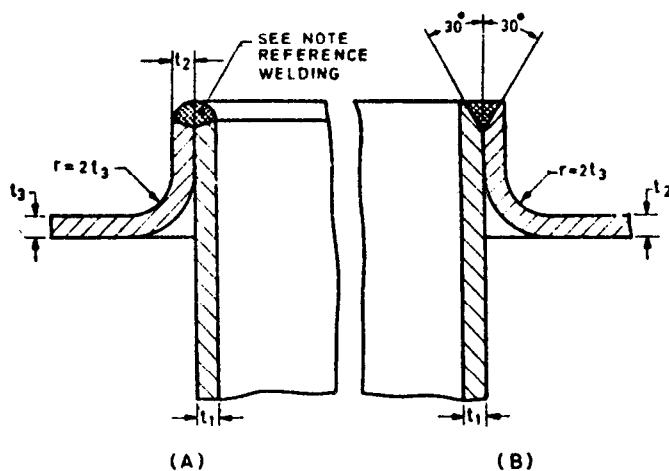
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16. (Page 198, Fig. G.12)
- Delete the words ' (Recommended for light duty vessels) '.
 - Substitute the following for the existing note:
 - Permitted for Class 3 vessels only.
 - The pad is not to be taken into account in calculating the reinforcement required.'
17. (Page 206, Fig. G.34) — Incorporate a fillet weld at the edge of lower plate.
18. [Page 207, Fig. G.36 (under consideration)] — Delete the figure.
19. [Page 209, Fig. G.40(B)] — Add the following note at the end and renumber the existing note as note (a):
- Permitted for Class 3 vessels only.'
20. (Page 209, Fig. G.41) — Add the following note at the end:
- Permitted for Class 3 vessels only.'
21. (Page 216, Fig. G.67)
- Substitute the following for the existing note (b):
 - Minimum distance between tubes = $2.5 t$
 - Add the following new note (f):
- A satisfactory weld procedure should be established to ensure that the throats at all sections (minimum leakage path) shall not be less than the minimum required.'
22. (Page 217, Fig. G.70) — Add the following new note (c):
- A satisfactory weld procedure should be established to ensure that the throats at all sections (minimum leakage path) shall not be less than the minimum required.'
23. (Page 218, Fig. G.71) — Add the following figure (D).



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24. (Page 218, Fig. G.72) — Substitute the following for the existing figure and the notes thereunder:



- t_1 is the thickness of the tube. It shall not be less than 2.5 mm.
- t_2 is the original plate thickness.
- t_2 is the plate thickness after bunging. t_2 should not be less than t_1 .
- It is uncommon for this detail to be used if the thickness of the tube or tube sheet exceeds 4 mm.
- Reference Welding

The detail shown in Fig. G.70 is suitable for welding by processes other than the metal arc process. A filler rod should be used if the tube wall thickness exceeds 1.5 mm when oxy-acetylene gas welding is employed and 2.9 mm when other suitable arc welding processes are used, such as atomic hydrogen or inert gas arc welding.

FIG. G.72 TUBE TO TUBE PLATE CONNECTIONS

25. (Pages 219 and 220, Fig. G.74, G.75, G.76, G.77, G.80, G.81, G.82) — Add the following additional note under all these figures:

'NOTE — If plate material is used for tube sheets, special attention should be given to examination of lamellar defects before and after welding.'

26. [Page 220, Fig. G.78, G.79, G.83(A), G.83(B)] — Add the following additional note for all these figures:

'NOTE — Hubs for butt welding to adjacent shell, head or other pressure parts shall not be machined from rolled plate and the component having the hub shall be forged to provide in the hub the full minimum tensile strength and elongation specified for the material in a direction parallel to the axis of the vessel. Proof of this shall be furnished by a tensile test specimen taken in this direction as close to the hub as possible.'